Labor Efficiency Lowers CRC Costs

Application | Maintenance | Site Management | **Component Rebuild** | Component Life Management | Safety | MARC Management

1.0 Introduction

Component rebuild costs are driven by two factors:

1. Parts consumed:
    - Parts consumption is the result of parts reuse and salvage capability.

2. Labor consumed.
    - Labor is consumed in processes, tooling, and asset utilization.

Most large mining customers utilize exchange components in order to minimize the impact of repairs/rebuilds on machine availability. Component exchange programs require “flat rates,” so that when improvements are made to shop processes (thereby increasing labor efficiency), the hours saved, reduce the cost of the rebuild and increase the profit margin.

In addition to creating a higher profit margin, the hours reduced through efficiency improvements become available to be used to handle additional work. Labor efficiency therefore has a compounded impact on both profit margins and shop capacity.

2.0 Best Practice Description

The basic principle behind improving labor efficiency is simply stated as identifying and eliminating waste. Waste is defined as “any activity that consumes resources but creates no value for the customer.”

Some waste is obvious, such as time spent “waiting” for something. This includes waiting for parts, waiting for access to tooling, forklifts, overhead cranes, and waiting for process delays caused by bottlenecks.

Other forms of waste are not so obvious, such as underutilized assets or using skilled employees to do unskilled work.

Part of the issue relates to culture. Long established shops with set procedures and processes often have wasteful habits that are simply “the way things have always been done.” In such an environment, it is hard to identify the waste and harder still to change the culture.

Some dealers have created strategic campaigns to raise awareness of the definition of waste, the cost of waste, and to involve all employees in a “War on Waste.”

The key success factors in improving labor efficiency are:

- Create detailed process maps for all shop areas.
- Train all employees on the definition and cost of waste.
- Engage all employees in the elimination of waste.
- Assign shop floor supervisors with responsibility to improve efficiencies in their work areas.

Often there are costs involved with effort to improve efficiency. The cost can come from improved tooling, facility design and layout, work flow processes, or staff changes. Where costs are involved, a benefit versus cost assessment should be analyzed.

There are numerous training materials available on managing efficiency by eliminating wasteful non-value added activities. Caterpillar has a waste elimination program for manufacturing plants titled *Caterpillar Production Systems* (CPS) (Form numbers YEGT 1751, 1752, 1753, and 1758). While the program is manufacturing-oriented, the principles also apply to a CRC environment. There are also numerous references available on the Internet.

3.0 Implementation Steps

Three generally accepted areas of change drive labor efficiency/ Continuous Improvement. Those are:

- Operational Changes. Identify and eliminate waste. 6 Sigma and Process Mapping are effective tools to for operational changes.
- Culture. Train and engage employees to drive changes from the bottom up. Create an attitude that makes change possible and improves the way we work.
- Management. A dedication to Continuous Improvement and measuring performance results.

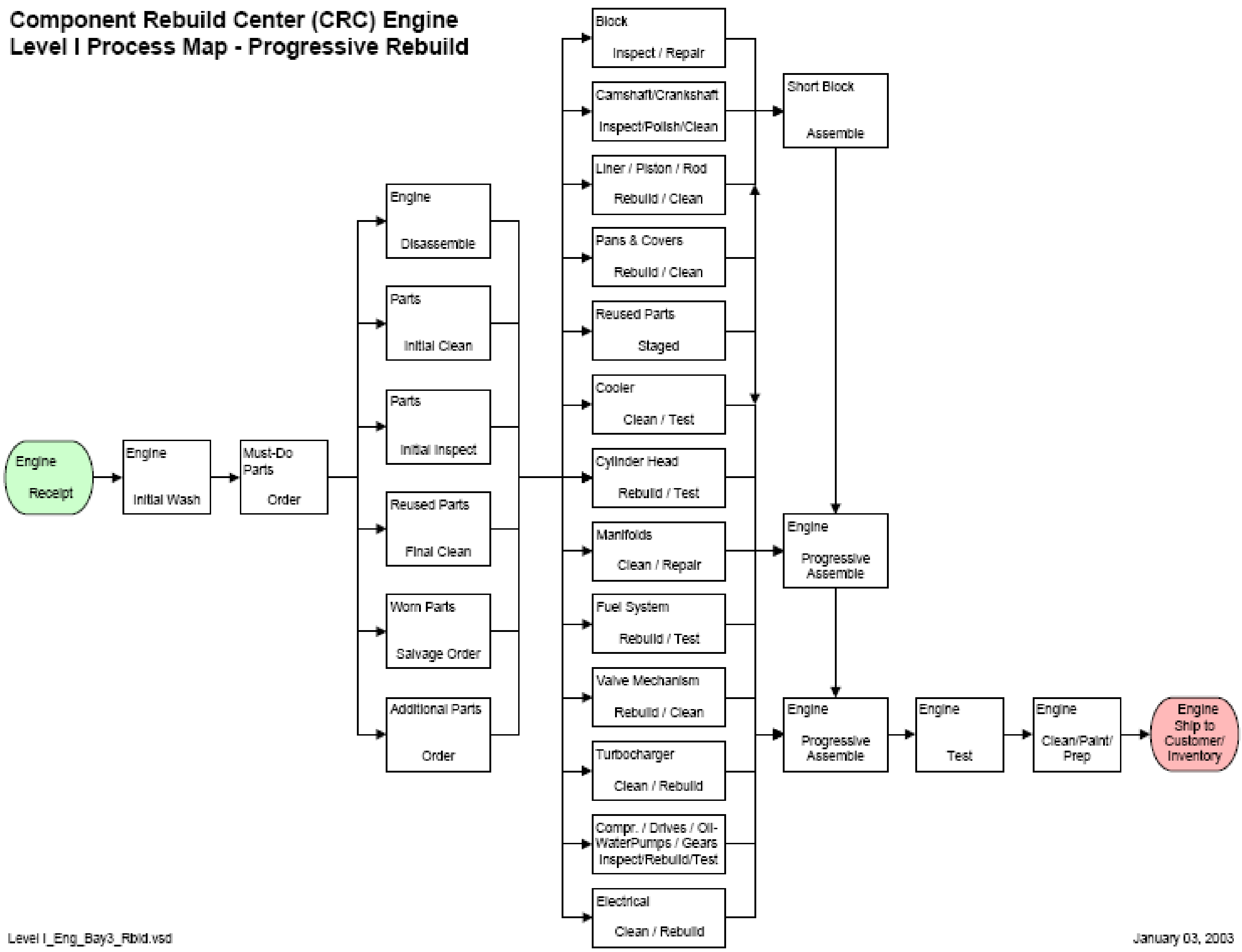
Common Sources of Waste

- Waiting
    - For parts
    - For tooling
    - For service literature or reference information
    - For customer W/O authorization
    - For process bottlenecks & interruptions
    - For Machining shop support
    - For Manpower resource
- Excess Motion
    - Excess movement in work station
    - Getting parts
    - Getting tools
    - Moving components and parts
    - Moving tool box
    - Looking for parts / tools in workbay
    - Power tools
- Asset utilization
    - Skilled mechanic doing unskilled labor
    - Excessive storage of in-process work
    - Poor safety – employee lost time
    - Workload variation
    - Tooling management, maintenance, & calibration
- Work done over
    - Service rework
    - Parts cleaning multiple times
    - Parts order and re-orders.
    - Moving components & parts in and out of staging
    - Dyno test more than once
    - Damaged or rusted parts

- Tooling efficiency
    - Proper cleaning equipment
    - Proper machining / salvage equipment
    - Component rebuild stands / rollover equipment

- Material handling – overhead cranes, jib cranes, slings, hooks, forklift
- Storage – staging areas and storage racks, baskets
- Cleanliness protection

The two most valuable management tools for labor efficiency improvement are process maps and time studies. These analysis tools will help identify work-in-process bottlenecks and time spent in each step of the rebuild process. It also helps to identify duplications within the process, such as repeated cleaning of parts, material handling duplications, and other non-value added activity.



Example of Level I Process Map

Below are examples of actual dealer efficiency improvement actions:

Labor Efficiency Issue	Solution
Parts order and re-orders causing delays	Standard "must-do" kits for common rebuilds Parts evaluated and ordered during component disassembly.

	Sub-components areas have common must-do parts cabinets. Parts delivered direct to work station
Improper tooling causing labor inefficiency	Machining shop buys improved high efficiency line bore and engine block resurface tooling New horizontal cabinent washer.
Parts / labor to recondition 3500 cylinder head exceeds price of Reman head group	Utilize Reman head groups as standard "Must Do" parts kit.
Mechanics spend too much time in assembly work station looking for parts and tools .	Standardize assembly area toolbox and toolbox layout. Develop organized assembly parts cart and standard baskets.

Example: Cooler assembly workstations:

- Efficiency enhancements in the station include a repair stand with easy access and a wall mounted, organized toolbox with all required tools for that workstation. The tooling stays in the work area and is dealer-owned. See Related Best Practices section.

4.0 Benefits

- Improved service profit margins – especially for flat rate rebuilds.
- Increased shop capacity and higher revenues
- Better component turnaround time – requires less Exchange inventory
- Better working environment
- Reduces need for additional shop space and manpower

The following example is actual dealer results:

CRC SAVE CALCULATION SPREADSHEET					
PERIOD	JULY – DEC 2002	JAN – JUNE 2003	JULY – DEC 2003	JAN – JUNE 2004	JULY – DEC 2004
TAT AVERAGE(CALENDAR DAYS)	41,8	46,2	26,5	23,8	19,8
LABOR SPENT FOR MAYOR COMPONENTS REPAIRED	147.455	161.675	170.508	102.694	93.832
MAYOR COMPONENTS REPAIRED	636	771	855	560	630
HRS/COMPONENT	231,8	↓ 209,7	↓ 198,5	↓ 183	↓ 148,9
SAVE IN LBR		17.039 Hrs	28.347	27.328	56,454
PRODUCTIVITY IMPROVEMENT (%)		9,5	14,3	21	35,8
TOTAL SAVINGS					129,168 hrs

5.0 Resources Required

- Management commitment to a culture change and Continuous Improvement process
- Employee training on waste and eliminating waste
- Project team (possibly 6Sigma)
- Cost versus benefit to improve shop tooling and processes.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

Component Turnaround Time	0906-4.5-1013
Using Cameras to Conduct CRC Process Reviews	0408-4.1-1117
CRC Hand Tool Management	0107-4.4-1057
Floor Mounted Cooler Assembly Fixtures	1006-4.4-1023

8.0 Acknowledgements

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